BE-02: Existing System Comparison

Intelligent Kanban System with an AI Assistant

Symbat Saparbay | Akbope Bakytkeldy

YouTrack: DEMO-21

1 Purpose

This document compares existing task-management systems against one consistent set of criteria in order to answer two questions:

1. Which parts of the proposed voice-to-Kanban workflow already exist in other products?
2. Which combination of capabilities is still missing, and therefore justifies building our system?

Every claim about another product is linked to a source (Section 8). Where a feature is not described in the sources we reviewed, the table says "Not found" rather than "No", because absence from documentation does not prove that the feature does not exist. Sources were checked on 21 October 2026; AI features in these products change quickly.

2 Selection of Systems

The systems were selected to cover the three groups our target users (employees, project managers, teachers and students) actually use or could use:

System	Group	Why it was selected
Trello	Kanban board (mainstream)	The reference Kanban tool; closest to our board user interface; has built-in AI features
Jira + Rovo	Team project management (mainstream)	Standard tool for software and project teams; AI assistant Rovo creates and updates work items
ClickUp + Brain	All-in-one work management (mainstream)	Combines boards with an AI notetaker that turns meeting action items into tasks, and voice "Talk to Text"
Todoist + Ramble	Task manager with voice capture	The most direct example of voice-to-task: Ramble turns speech into several tasks with dates, priorities and assignees
Hypertask	AI-native project board	A new Kanban board shared by people and AI agents, with AI chat and AI task writing; identified in the team's initial review

Screened but excluded. The team's initial review (DEMO-21, first draft) also covered *Vibe Kanban*, *Formic* and *KaibanJS*. They were excluded from the main comparison because they target software developers and coding agents (running agents on code, decomposing coding goals, or a developer framework) rather than general teams. Formic's decomposition of a goal into 3–8 subtasks remains a useful reference for our multi-task extraction.

3 Comparison Criteria

The nine criteria follow directly from the project requirements (product requirements document and BE-03):

ID	Criterion	Why it matters for our project
C1	Team Kanban board with roles	Core product: shared boards, members and permissions
C2	Voice input that produces tasks	Main input channel of our assistant
C3	Several tasks from one message or meeting	Supervisor cases 2 and 6: several assignments in one utterance, meeting outcomes
C4	Changing existing tasks with natural language	Cases 3 and 4: change deadline, priority, status, assignee
C5	Answers based on real project data	Case 5: overdue tasks, project summary without invented tasks
C6	User review before AI changes are saved	Our core safety rule: AI only proposes
C7	Kazakh and mixed Kazakh–Russian speech	Case 7; our main market
C8	Data hosting and private AI mode	Voice recordings are sensitive; some organisations cannot send data to external clouds
C9	Cost of AI features	Affordability for student and small teams

4 Comparison Table

Criterion	Trello	Jira + Rovo	ClickUp + Brain	Todoist + Ramble	Hypertask	Our system (planned)
C1 Team Kanban board with roles	**Yes** Boards and workspaces[1]	**Yes** Boards and project permissions[3]	**Yes** Board view in workspaces[6]	**Partial** Board view, teams; task manager first[8]	**Yes** Kanban board shared by people and agents[10]	**Yes** Workspaces, projects, boards, roles
C2 Voice input that produces tasks	**Not found** No voice input in reviewed sources[1,2]	**Not found** Not mentioned on the Jira AI page[3]	**Yes** "Talk to Text" converts ideas into tasks[7]	**Yes** Ramble turns speech into tasks[8,9]	**Partial** Voice dictation reported in the team's review of product material; not shown on the current landing page[10]	**Yes** Browser recording, self-hosted STT
C3 Several tasks from one message or meeting	**Partial** AI-generated board from a described goal (text)[2]	**Partial** Captures work from Confluence, Slack, email; breaks work into smaller tasks[3]	**Yes** AI Notetaker turns meeting action items into assigned tasks[5]	**Yes** Several tasks per session (pause between items)[9]	**Not found** No extraction of several tasks from speech found[10]	**Yes** Several actions per utterance; meeting summary to task drafts
C4 Changing existing tasks with natural language	**Not found**	**Yes** Update work items from Rovo chat[3]	**Partial** Autopilot updates priorities and assignments by triggers, not by command[6]	**No** Edits only inside the current session preview[9]	**Partial** AI chat manages the board[10]	**Yes** Update, status, assignee, deadline, priority by voice or text
C5 Answers based on real project data	**Partial** Card content summarisation[2]	**Partial** Work item summaries and search[3]	**Partial** AI Search & Ask over tasks and docs (Enterprise AI)[6]	**Not found**	**Partial** AI chat over the board; grounding not documented[10]	**Yes** Search and summary run as database queries; no invented tasks
C6 User review before AI changes	**Not found**	**Not found**	**Not found** Review step before tasks are created not described[5]	**Yes** Live preview; tasks added with "Add tasks"[9]	**Partial** Approval gates for agent work (team review)[10]	**Yes** Every change requires confirmation; idempotent confirm
C7 Kazakh and mixed speech	**Not found**	**Not found**	**Not found** Languages not listed[5,7]	**No** 40+ languages incl. Russian; Kazakh not listed[9]	**Not found**	**Yes** Kazakh model + language routing for mixed speech
C8 Data hosting and private AI mode	**No** Cloud service only	**Partial** Full AI needs Atlassian Cloud; Data Center only via connectors[4]	**No** Cloud service	**No** Cloud service	**Partial** Hosted; bring-your-own-key plan[10]	**Yes** Voice processed on own server; optional local LLM
C9 Cost of AI features	**Yes** AI included in Standard and Premium plans[2]	**Partial** Included in Cloud plans with credit limits (25/70/150 per user per month); overage $0.01 per credit[4]	**Partial** Brain $9, Everything AI (with Notetaker) $28 per user per month, billed for every member[6]	**Partial** 10 sessions per month on Free/Beginner; unlimited on Pro/Business[9]	**Partial** Free starter allowance; $8 (own keys) or $16 per user per month[10]	**Yes** Estimated ≈$0.005 per command (BE-03)

Yes documented in the sources; **Partial** partly or with limitations; **No** documented as not supported; **Not found** not found in the reviewed sources. Superscripts refer to Section 8.

5 Strengths and Limitations of Each System

System	Strengths	Limitations for our users
Trello	Simple, familiar Kanban board; AI content generation and AI-generated boards now included in Standard plans	No voice input or command-based task editing in the reviewed sources; AI works on card text, not on team workflows; cloud only
Jira + Rovo	Strong AI: captures work from several sources, breaks work down, updates items from chat, summarises; mature permissions	Complex for students and small teams; full AI only in Atlassian Cloud; credit limits; no voice or Kazakh support found
ClickUp + Brain	Closest to our meeting scenario: AI Notetaker turns action items into assigned tasks; voice "Talk to Text"	AI is a paid add-on billed for every workspace member; review before task creation not described; languages not listed; cloud only
Todoist + Ramble	Best voice-to-task experience: several tasks per session, dates, priorities, assignees, live preview before adding	Personal task manager first; cannot change existing tasks by voice; Kazakh not supported; free plan limited to 10 sessions per month
Hypertask	Board designed for people and AI agents together; AI chat and triage; bring-your-own-key option	Young product; most information comes from the vendor's own website; no multi-task extraction from speech found; Kazakh not documented

6 Gap Analysis and Justification

Each capability of our system already exists *somewhere*: Todoist extracts several tasks from speech with a preview, ClickUp turns meeting action items into tasks, Jira's Rovo updates work items from chat. The gap is in the **combination** and in the **local context**:

1. **No reviewed system covers C2–C6 together.** Todoist creates tasks by voice but cannot change existing ones; Jira changes existing items but has no voice input; ClickUp converts meetings into tasks but does not describe a review step.
2. **Kazakh and mixed Kazakh–Russian speech (C7) is not supported or not documented** in any of the five systems. Todoist explicitly lists Russian but not Kazakh.
3. **Voice data stays in the cloud (C8)** in all mainstream systems. Our design processes voice on our own server and offers a local LLM mode.
4. **Safety rule for team data.** Only Todoist documents a review step, and only for new personal tasks. Our system requires confirmation for every change (create, update, status, assignment) and checks the user's permissions.

Conclusion. The proposed system is justified not by a single new feature, but by bringing together voice and text commands for the full task life cycle on a team Kanban board, with mandatory confirmation, answers grounded in real project data, and support for Kazakh, Russian, English and mixed speech.

6.1 Ideas Adopted from Existing Systems

- From **Todoist Ramble**: live preview while the user speaks; voice corrections such as "remove that"; pausing between items to separate tasks.
- From **ClickUp AI Notetaker**: meeting recording → action items → task drafts with assignees.
- From **Jira Rovo**: breaking larger work into smaller tasks; summaries with status, blockers and next steps.

- From **Formic** (excluded tool): decomposition of a goal into a bounded number of subtasks.

7 Limitations of This Comparison

- Most sources are vendor pages and documentation; they describe intended functionality, not measured quality. Third-party reviews are used only for pricing.
- The systems were not tested hands-on with Kazakh or mixed speech. "Not found" means not documented, not proven absent.
- AI features and prices change frequently; the comparison reflects sources checked on 21 October 2026 and should be re-checked before the final defence.

8 References

[1] Trello. Pricing and plans. https://trello.com/pricing

[2] PricingSaaS. Trello: AI features moved to Standard plan (capture of 3 August 2026). https://pricingsaas.com/companies/trello/diffs/2026W32

[3] Atlassian. AI in Jira (Rovo). https://www.atlassian.com/software/jira/ai

[4] Atlassian. Rovo licensing and credits. https://www.atlassian.com/licensing/rovo

[5] ClickUp. AI Notetaker. https://clickup.com/brain/ai-notetaker

[6] eesel AI. ClickUp Brain review: features, pricing and limits (2026; written by a competitor, used for pricing). https://eesel.ai/blog/clickup-brain

[7] ClickUp. Talk to Text. https://clickup.com/p/talk-to-text/capturing-ideas

[8] TechCrunch (21 January 2026). Todoist's app now lets you add tasks by speaking to its AI. https://techcrunch.com/2026/01/21/todoists-app-now-lets-you-add-tasks-to-your-to-do-list-by-speaking-to-its-ai/

[9] Todoist Help Center. Dictate to add tasks with Ramble. https://todoist.com/help/articles/215532676695580

[10] Hypertask. Product website and pricing. https://hypertask.ai